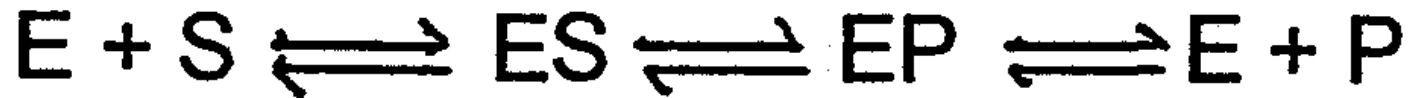


Enzymes

Introduction

- ▶ *All reactions in the body are mediated by enzymes, thus enzymes direct all metabolic events.*
- ▶ *Enzymes :are biological catalysts produced by living cells but independent of these cells for their action.They can work outside these cells.*
- ▶ *Substrate :is the reactant that enzyme act upon it and converted to the product.*
- ▶ *Product :substance that formed at the end of the chemical reaction.*



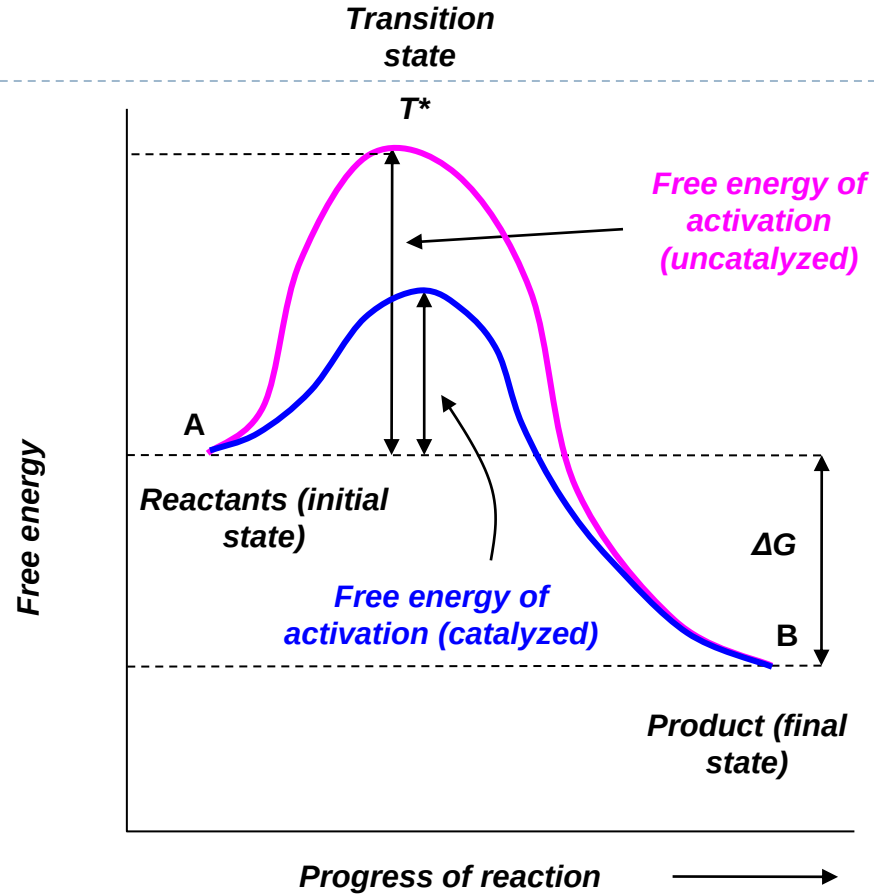
E = enzyme P = product

S = substrate

ES = enzyme-substrate complex

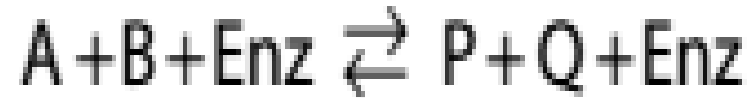
EP = enzyme-product complex

-
- ▶ For a reaction to proceed from starting material to product, the chemical transformations of bond-making and bond-breaking require a minimal threshold amount of energy, termed **activation energy**. Generally, a catalyst serves to lower the activation energy of a particular reaction.



What to know about the enzymes:

- ▶ *Virtually all enzymes are proteins, however, a few examples of RNAs show enzymatic activity; they are called ribozymes.*
- ▶ *They are very sensitive to slight changes in temperature and pH.*
- ▶ *Enzymes accelerate chemical reactions without changing the equilibrium point.*
- ▶ *They are highly specific in their action*
- ▶ *They function in minute amounts and remain unchanged chemically by the end of the reaction.*



$$K_{\text{eq}} = \frac{[P][Q][\text{Enz}]}{[A][B][\text{Enz}]}$$

$$K_{\text{eq}} = \frac{[P][Q]}{[A][B]}$$

Enzymes therefore have no effect on K_{eq} .

No change in the equilibrium of the reaction

Enzyme terminology

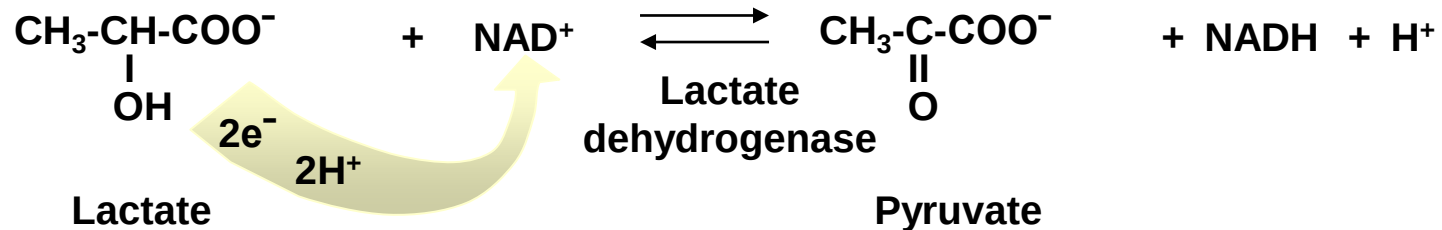
- ▶ **Recommended name (common):**
- ▶ Hydrolyzing enzymes are usually named by adding the suffix -ase to the name of the substrate e.g. amylase, lactase and lipase which hydrolyze starch, lactose and triacylglycerol respectively.
- ▶ Other enzymes are named by adding their function to the name of the substrate. Thus lactate dehydrogenase removes hydrogen from lactic acid and alanine transaminase transfers amino group from alanine.
- ▶ **Trivial name:** Some enzymes retain their original name, which give no hint of the associated enzymatic reaction e.g. pepsin and trypsin.

CLASSIFICATION OF ENZYMES

- ▶ According to the type of chemical reaction catalyzed, enzymes are classified into 6 major classes:
 1. Oxidoreductases
 2. Transferases
 3. Hydrolases
 4. Lyases
 5. Isomerases
 6. Ligases

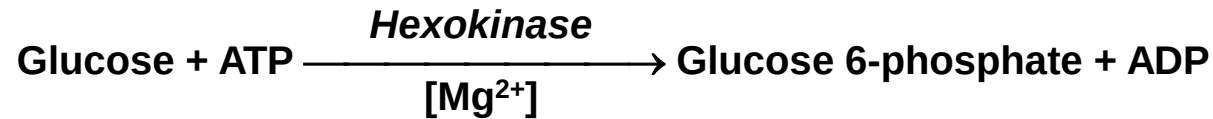
CLASS I: OXIDOREDUCTASES

- ▶ These are enzymes that catalyze oxidation-reduction reactions. The term **oxidation** means the **addition of oxygen** (e.g., addition of oxygen to hydrogen, forming water), the term **reduction** means the **removal of hydrogen** or the **positive increase** in valence (e.g., removal of electrons from ferrous iron, forming ferric iron).



CLASS II: TRANSFERASES

- ▶ Transferases are enzymes that catalyze the transfer of a chemical group from one molecule, the “donor”, to another, the “acceptor” According to the group transferred, they are classified into subclasses.



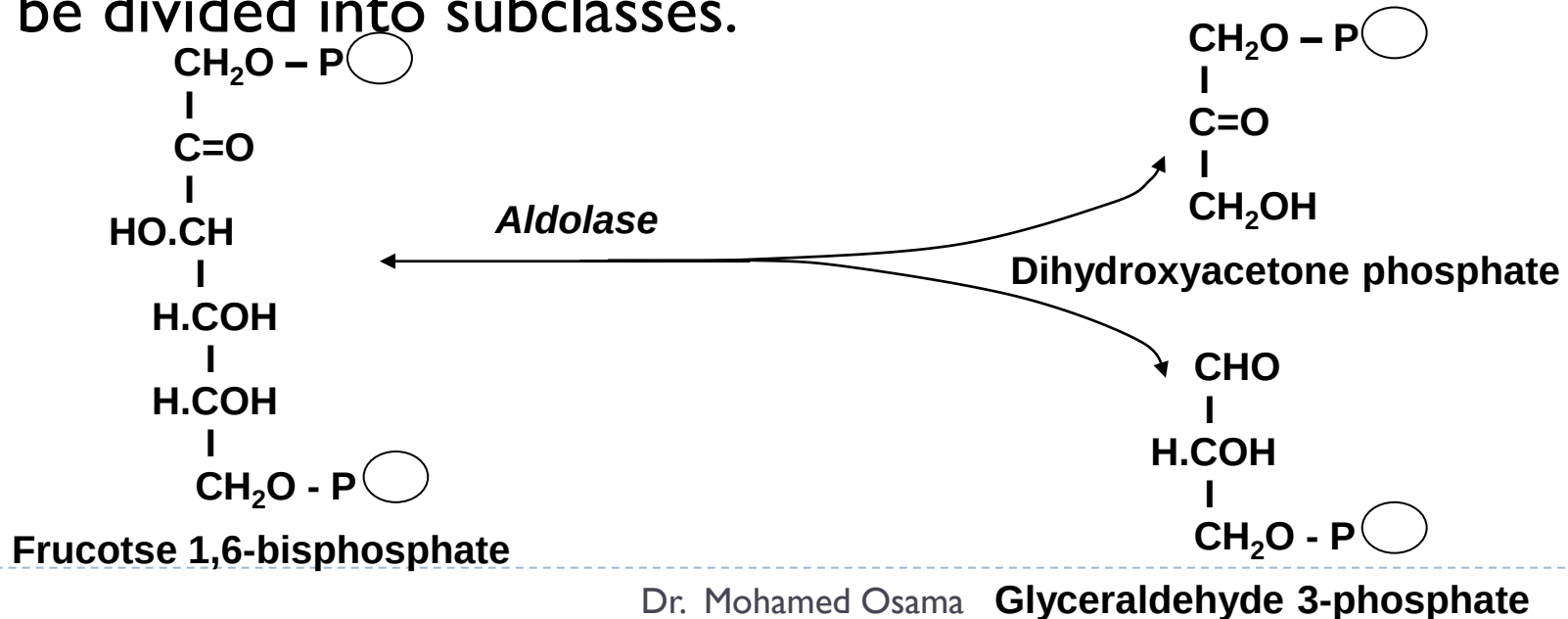
III: HYDROLASES:

- ▶ Hydrolases are enzymes that catalyze the reaction of hydrolysis, i.e., the breakdown of substrate by the addition of the elements of water across the bond broken. According to the bond cleaved, hydrolases classified into subclasses.



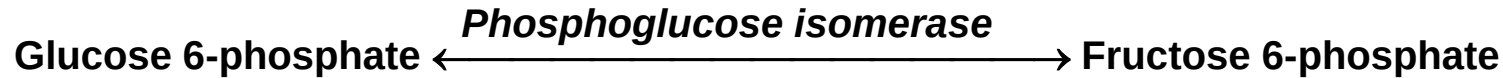
CLASS IV: LYASES

- ▶ Lyases, also known as splitting enzymes, are enzymes that catalyze the splitting of their substrate without hydrolysis, oxidation, or reduction. According to the group split, lyases can be divided into subclasses.



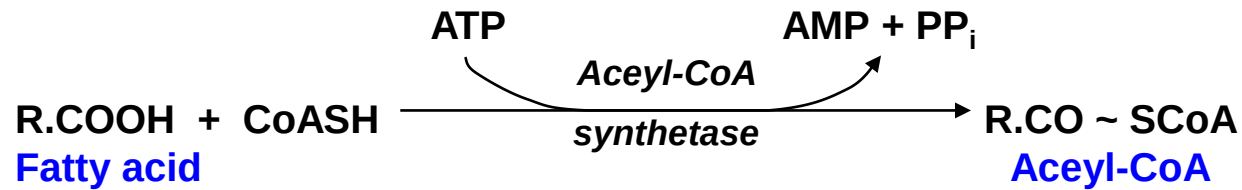
CLASS V: ISOMERASES

- ▶ Isomerases are enzymes that catalyze the interconversion of 2 isomers. According to the type of isomerism, they are divided into subclasses.



CLASS VI: LIGASES

- ▶ Ligases are enzymes that catalyze the binding of 2 molecules associated with the hydrolysis of high-energy phosphate bond, usually in ATP.

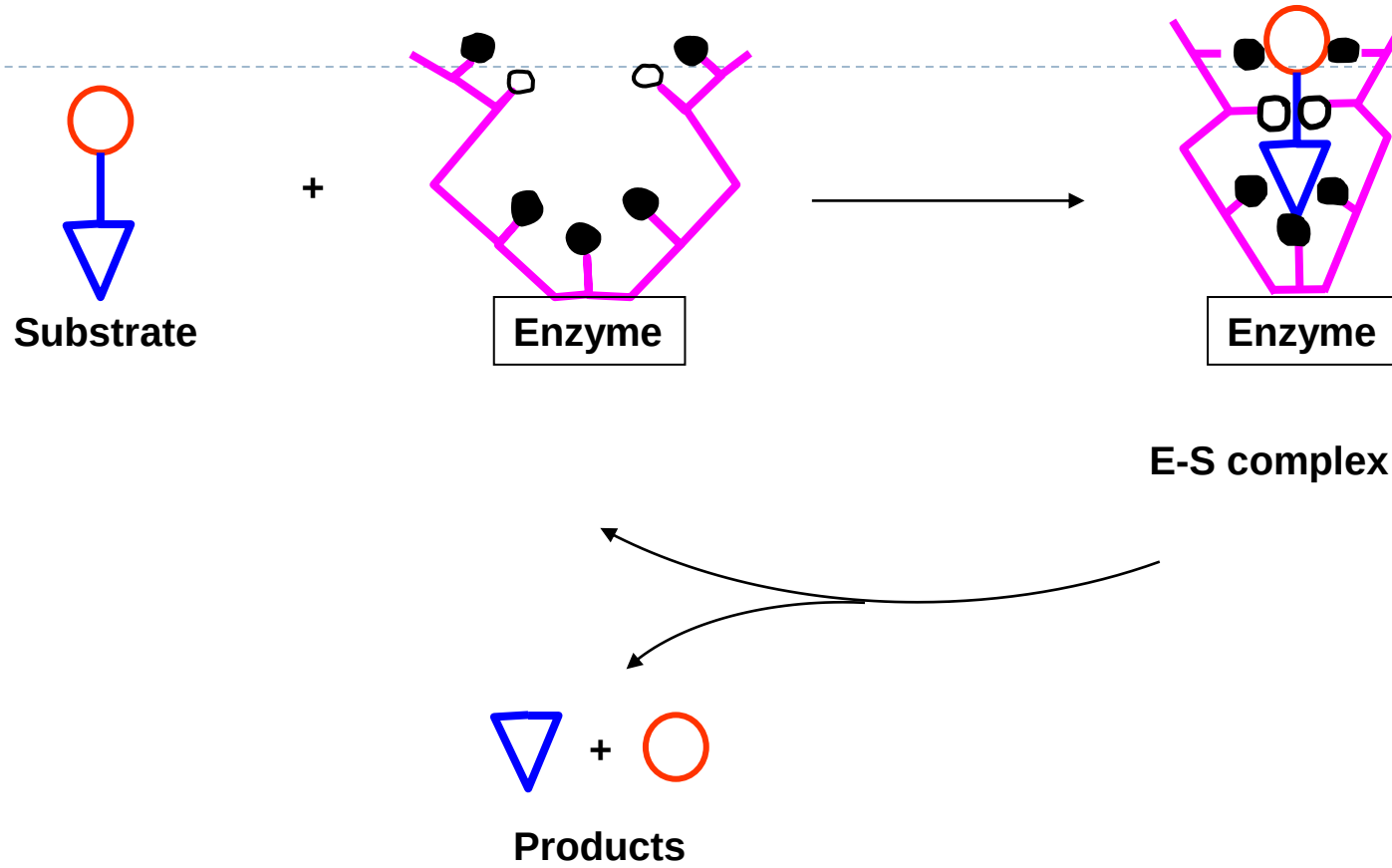


Enzyme structure

- ▶ **apoenzyme** - the protein portion of a holoenzyme minus the co-enzyme.
- ▶ **holoenzyme** - a complete, catalytically active enzyme including all co-factors.
- ▶ **co-factor** - organic or inorganic molecules that are required by some enzymes for activity. These include Mg^{2+} , Fe^{2+} , Zn^{2+} and larger molecules termed co-enzymes like nicotinamide adenine dinucleotide (NAD^+), coenzyme A, and many vitamins.
- ▶ **prosthetic group** - a metal or other co-enzyme covalently bound to an enzyme

ACTIVE SITE:

- ▶ Enzyme molecules contain a special pocket or cleft ,which is concerned with the binding of the substrate and the process of catalysis. This cleft is called the catalytic site or the active site. This small area consists of certain chemical groups [such as —OH, —SH, —NH₃, —COO⁻], found in the side chains of the amino acid residues of the enzyme protein. Some of these groups are concerned with the binding of the substrate , forming an enzyme substrate complex (E-S complex), they determine the specificity of the enzyme. Others activate some bonds in the substrate, lowering energy of activation and allowing the reaction to proceed rapidly.



-
- ▶ It was formally suggested that the catalytic site is a rigid area on the enzyme surface and that the substrate binds to the enzyme like “a key into its lock”. Now there is evidence that the catalytic site is a flexible area. The presence of the substrate induces a conformational change in the enzyme that allows the binding of the substrate and catalysis to occur. This means that we get “induced fit” of the substrate on the surface of the enzyme.

Enzyme Kinetics

- ▶ The study of the reaction rates of enzyme catalyzed reactions is called enzyme kinetics.
- ▶ Important for understanding how physiologic stresses such as anoxia, metabolic changes(acidosis or alkalosis),toxins, and pharmacologic agents effect.

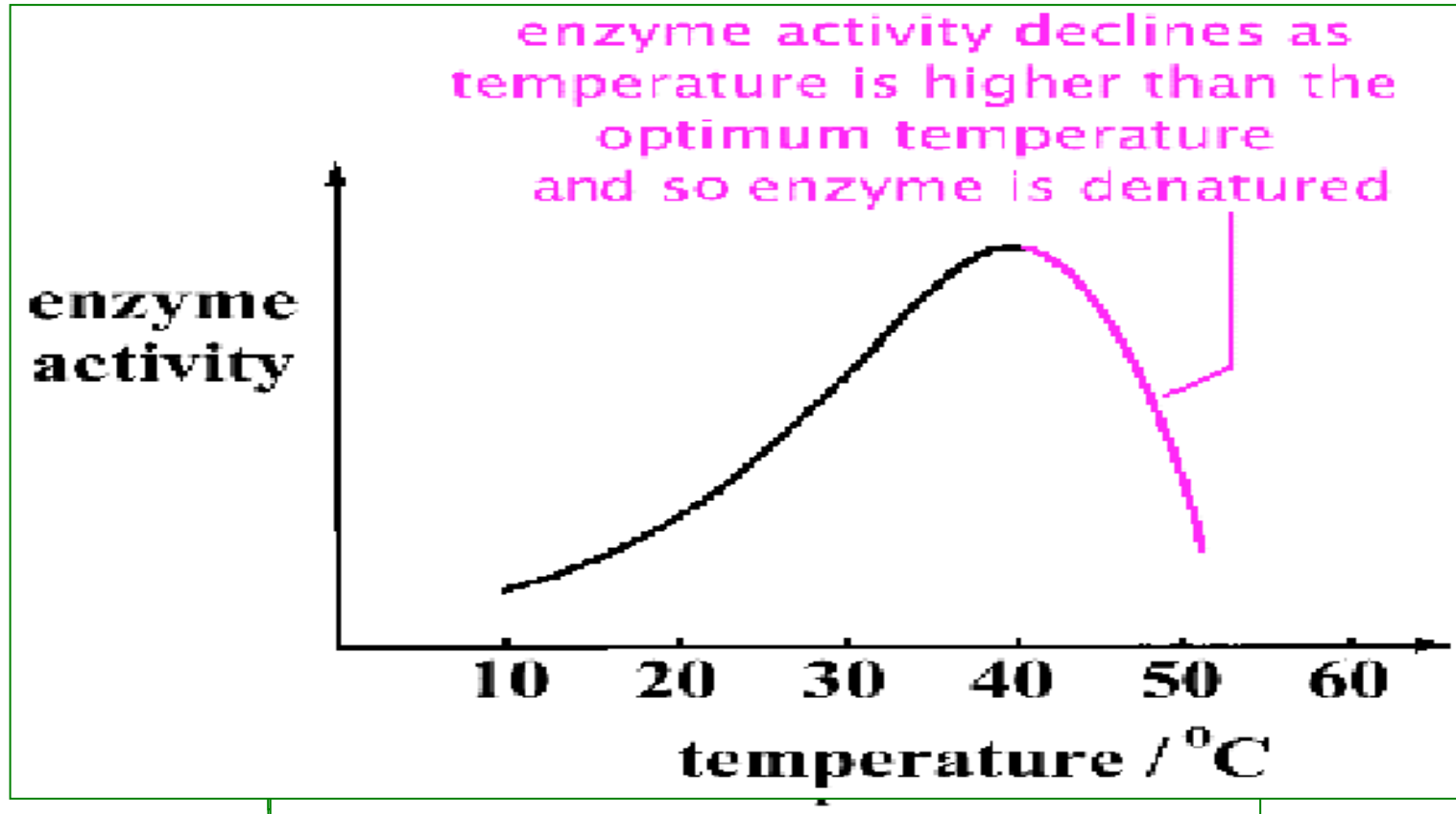
The rate of a reaction is influenced by several factors

- ▶ environmental conditions (like temperature and pH)
- ▶ concentrations of enzymes, substrates and products
- ▶ Effectors like activators and inhibitors

1. Temperature

- ▶ At first increase of V is due to the increased number of molecules having sufficient energy to form the products of the reaction. This is until a maximum velocity (V_{max}) is reached.
- Optimum temp is the temp that gives V_{max}
- ▶ Decrease of velocity with further elevation of temp causes temp. induced denaturation of the enzyme.

Effect of temperature on enzyme activity

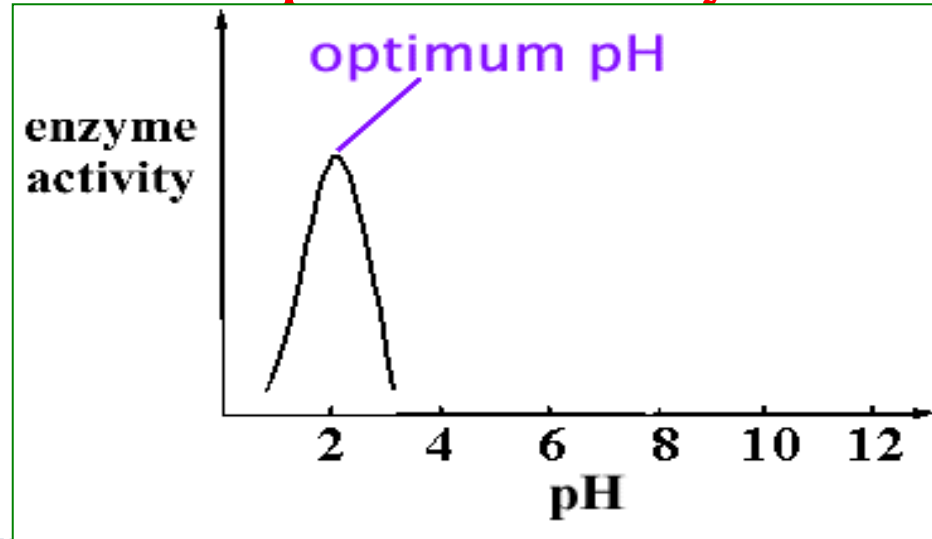


2- pH

- Effect of pH is on the ionization of the amino acid side chains involved in catalysis (active site)
- ▶ pH also affects the ionization of the substrate
- ▶ Optimum pH varies with different enzymes e.g., pepsin has an optimum pH of about 2, whereas most enzymes are designed to work at neutral pH (plasma pH 7.35 -7.45).

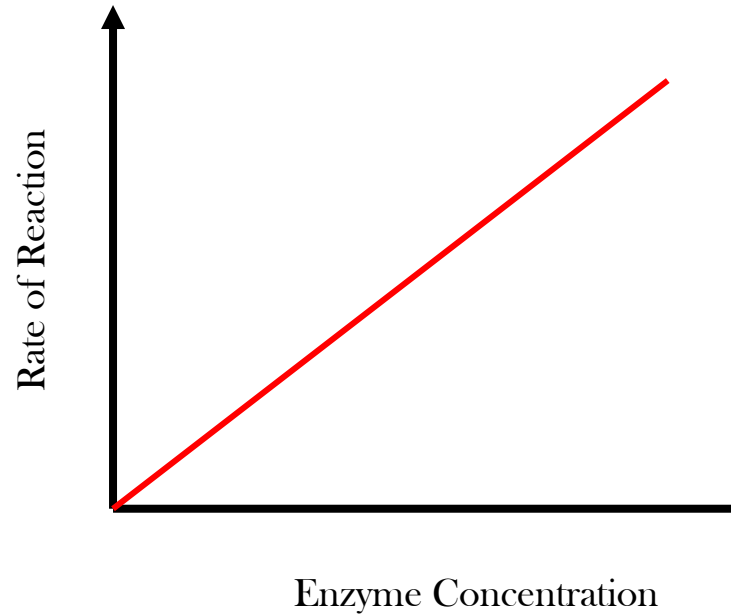
Effect of pH on enzyme activity

- ▶ Optimum pH : the pH value at which the enzyme has the highest activity
- ▶ pH higher or lower than optimum pH can inactivate the enzyme. Too extreme pH can destroy the enzymes



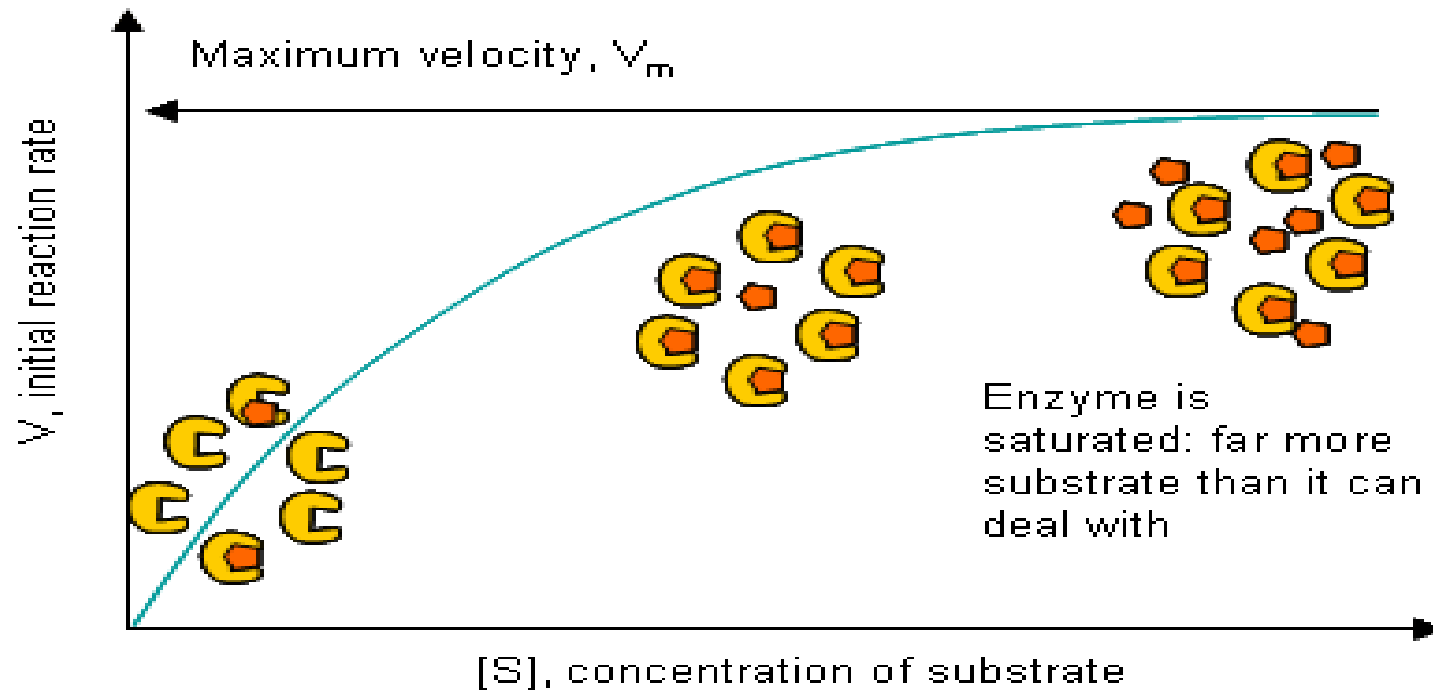
3. Enzyme Concentration

- Increasing enzyme concentration increases enzyme activity.

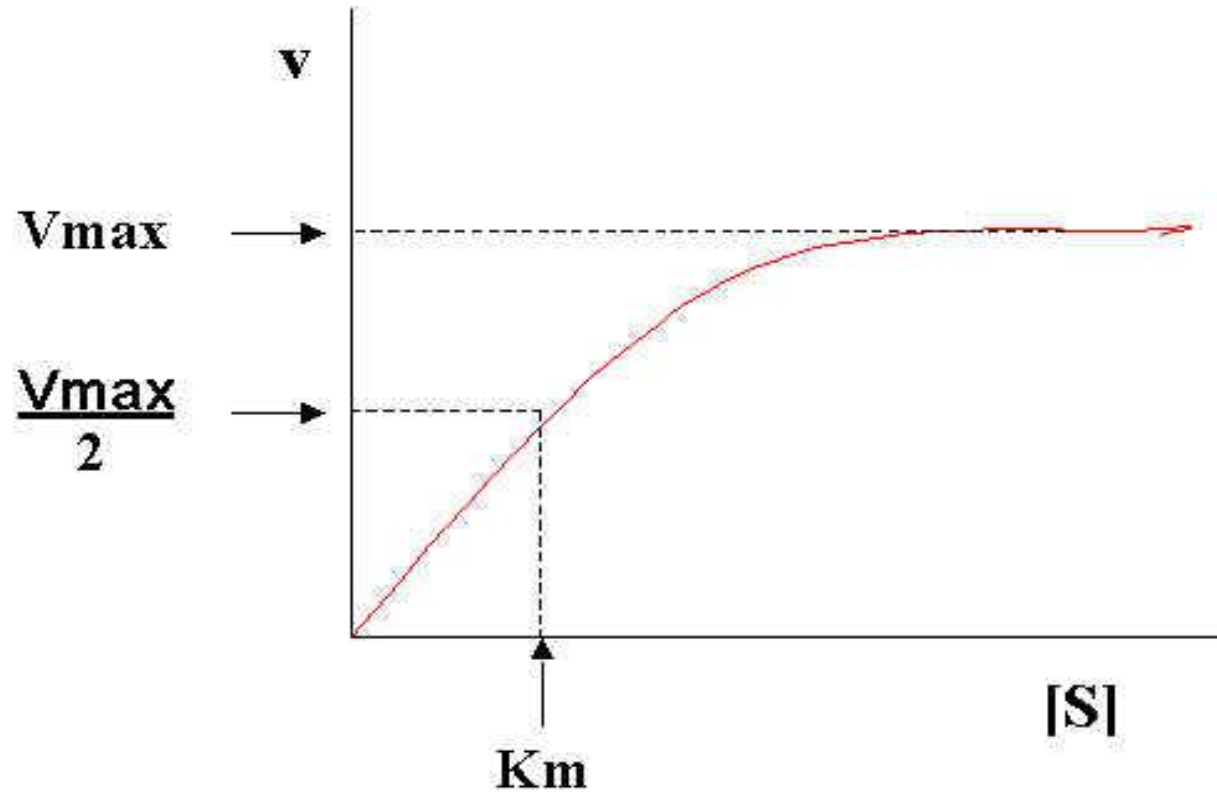


4. Concentration of substrate

increasing substrate concentration increases reaction rate until saturation of enzyme, then the reaction rate becomes constant.



K_m



K_m

- ▶ is the substrate concentration at half V_{max} .
- ▶ It is called Michaelis-Menten constant.
- ▶ K_m is inversely proportional to affinity of enzyme for substrate.
- ▶ **High** K_m means low affinity towards substrate
- ▶ **Low** K_m means high affinity towards substrate

5.Effecters

A substance that can binds to an enzyme and affects the enzyme activity is called an effector.

- ▶ **Activators** (positive effectors)
- ▶ **inhibitors** (negative effectors)

Inhibition of enzymes activity

- ▶ **Inhibitors are**
- ▶ Any molecule that binds to an enzyme and reduces enzyme activity.
- ▶ Inhibitors are classified into:
- ▶ **1- Irreversible inhibitors:** bind covalently (strongly) to the enzyme.
- ▶ **2- Reversible inhibitors:** Bind non-covalently (weakly) to the enzyme.

1- Irreversible Inhibitors

- ▶ Binding of the inhibitor to the active site blocks reaction. Enzyme is damaged permanently.
- ▶ Involves covalent modification at the active site.
- ▶ Irreversible inhibitors **are considered to be poisons** include:
 - ▶ **Arsenic**
 - ▶ **Snake venom**
 - ▶ **Nerve gas**

2- Reversible inhibitors

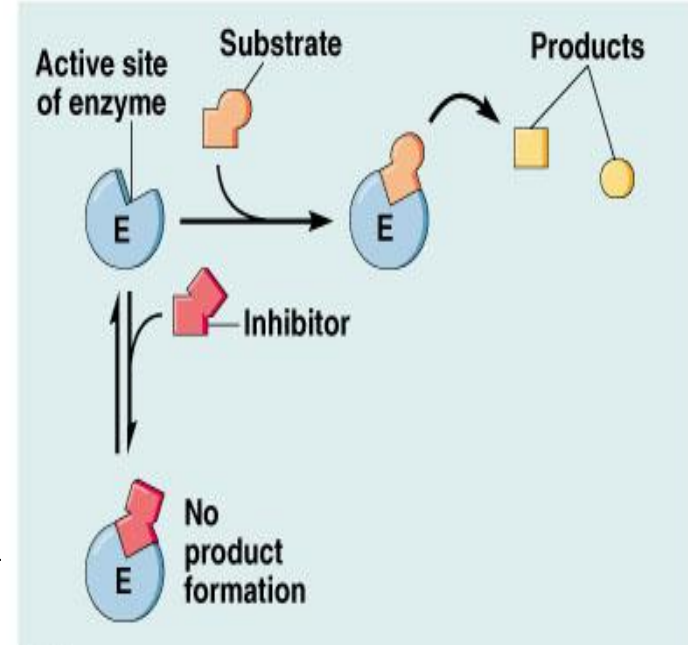
Can be

A- competitive inhibitors

B- noncompetitive inhibitors

A-Competitive Inhibition

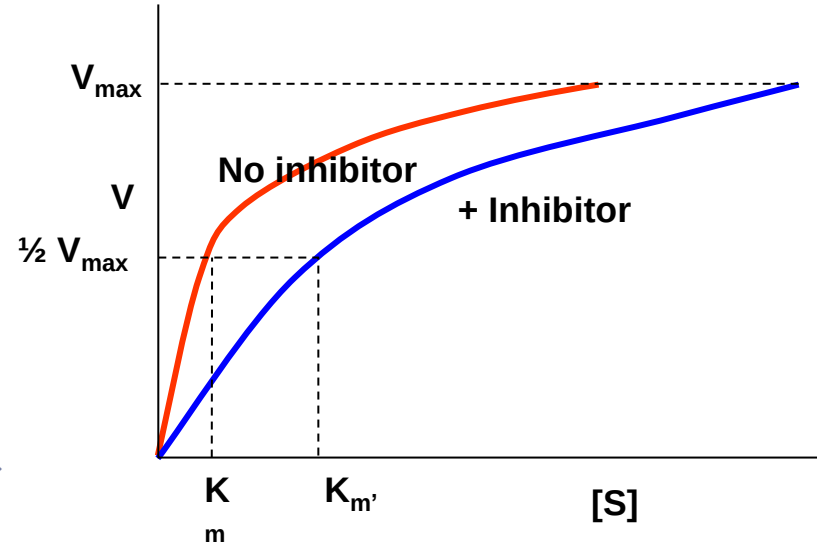
- ▶ has a structure like substrate
- ▶ Binds to active site of enzyme
- ▶ Prevents substrate from binding
- ▶ Can be released by increasing substrate concentration
- ▶ The degree of inhibition depend also on the affinity to the enzyme for both the inhibitor and the substrate.



▶ If we plot the velocity of the reaction as a function of the substrate concentration in the presence of a fixed amount of inhibitor, we get the curves shown.

▶ In the presence of the inhibitor, a higher substrate concentration is needed to reach a certain reaction rate; the apparent K_m , K_m' , is increased.

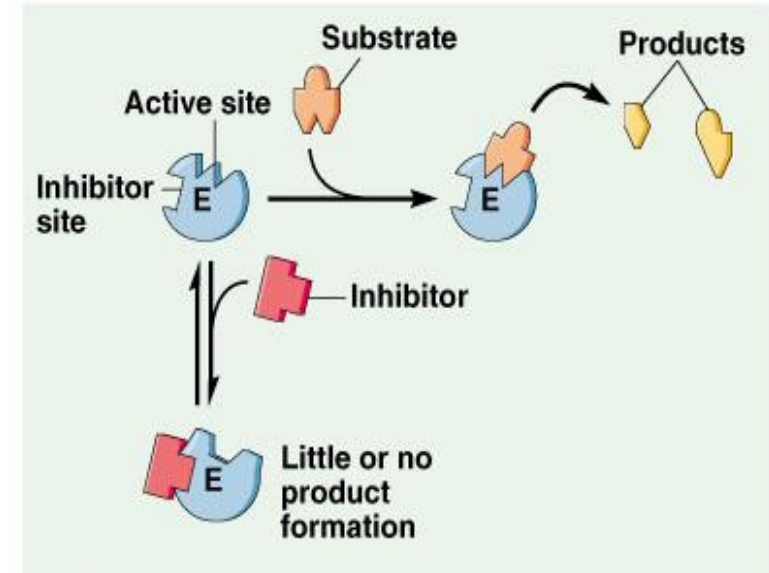
▶ However, at higher substrate concentrations, the inhibitor is completely displaced and the same V_{max} can be reached.

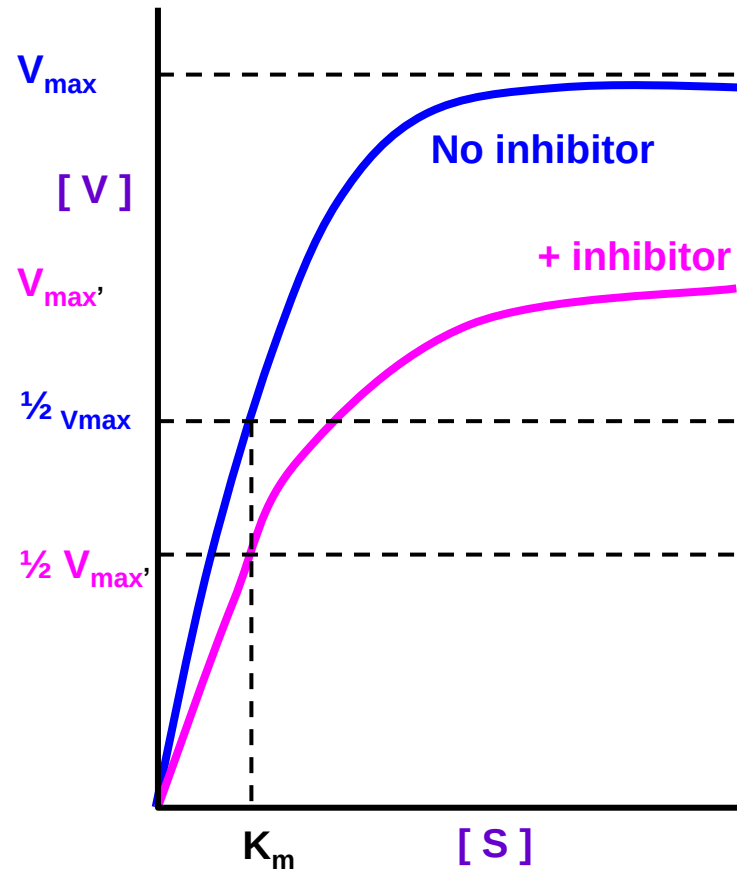


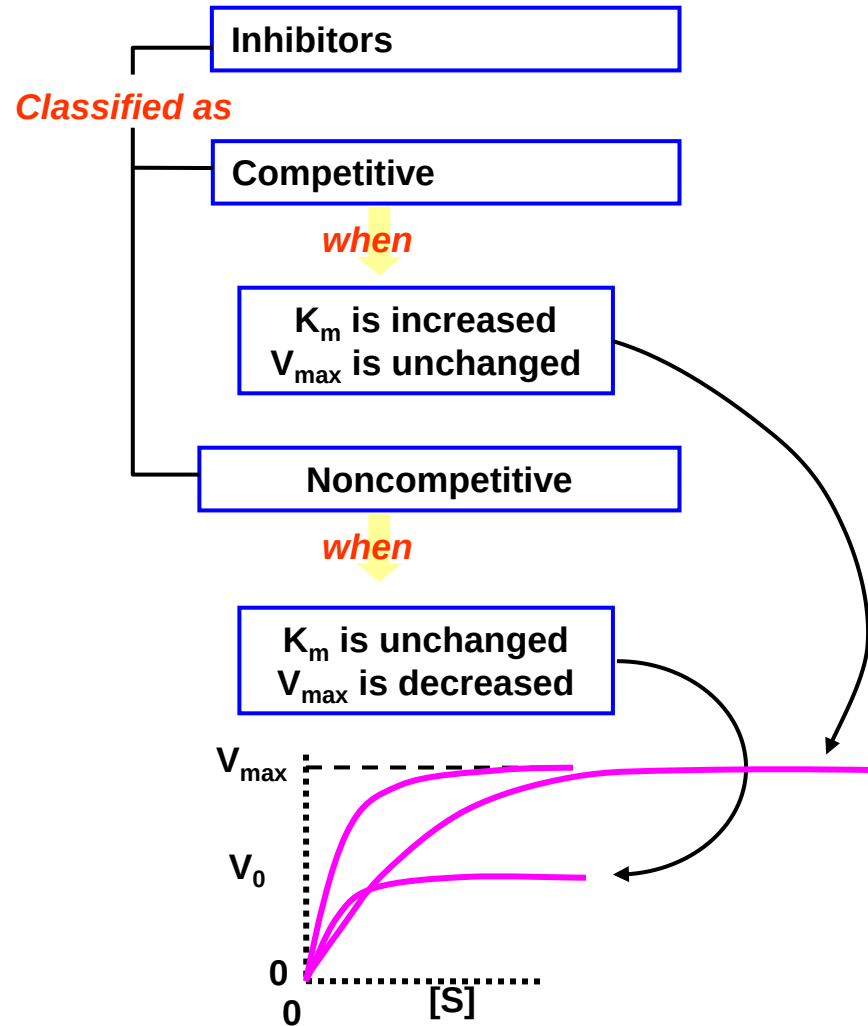
B - Noncompetitive Inhibition

A non- competitive inhibitor

- ▶ Does not have a structure like substrate
- ▶ Binds to the enzyme but not active site
- ▶ Changes the shape of enzyme and active site
- ▶ Substrate cannot fit altered active site
- ▶ No reaction occurs
- ▶ Effect is not reversed by adding substrate







REGULATION OF ENZYME ACTIVITY

- ▶ **Controlling Enzyme Activity**
- ▶ Enzymes are very efficient at what they do
- ▶ Because of this they need to be carefully controlled
- ▶ The cell also has to be able to regulate how much product is produced from a reaction.
- ▶ The activity of enzymes has to be regulated, so that they function at the proper time and place.

Some ways used to regulate enzyme activity are

1. Controlling the amount of enzyme present
2. Allosteric regulation
3. Feedback regulation
4. Produce enzyme in an inactive form (zymogen or proenzyme)
5. covalent modification

1.Regulation of enzyme quantity

- ▶ Rate of synthesis and breakdown of enzyme determine enzyme quantity.

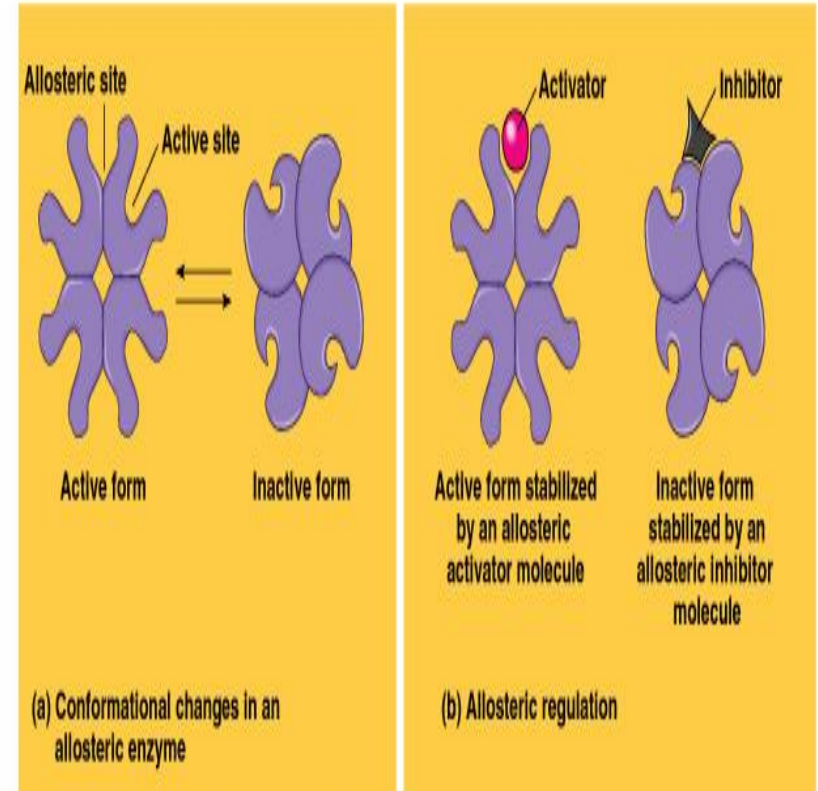
The quantity of an enzyme may be:

- ▶ increased by raising its rate of synthesis (induction)
- ▶ or decreased by lowering its rate of synthesis (repression).
- ▶ Enzyme synthesis is controlled by regulating gene transcription and translation.
- ▶ Takes hours or days to show full effect, It is used for long-term regulation.

2. Allosteric regulation

- ▶ An allosteric enzyme has another binding site.
- ▶ Effector molecules change the affinity of the enzyme for the substrate
 - ▶ Some effectors speed up enzyme action (positive effectors)
 - ▶ Some effectors slow enzyme action (negative effectors)

Allosteric effector could be products, substrate, and so on.



©1999 Addison Wesley Longman, Inc.

-
- They bind to certain chemical groups at a certain area on the surface of the enzyme, called the “**allosteric site**”, resulting in conformational changes that lead to changes in the arrangement of the chemical group at the catalytic site. ▶
 - Enzymes regulated by allosteric modifiers are called allosteric enzymes. They are usually **oligomeric**. ▶

- If the catalytic site becomes more suitable to bind with and ▶

activates the substrate, the substance causing this change is called
“positive effector” or **allosteric activator**.

-On the other hand, if the catalytic site becomes less suitable to ▶
bind with and activate the substrate, the substance causing this
change is called “negative effectors ” or **allosteric inhibitor**.

Allosteric enzymes

often

- Are usually composed of multiple subunits
- Catalyze a rate limiting reaction
- Bind substrate cooperatively
- Show a sigmoid curve when v_o is plotted against $[S]$

Bind allosteric effectors other than the substrate

which leads to

Activation

or

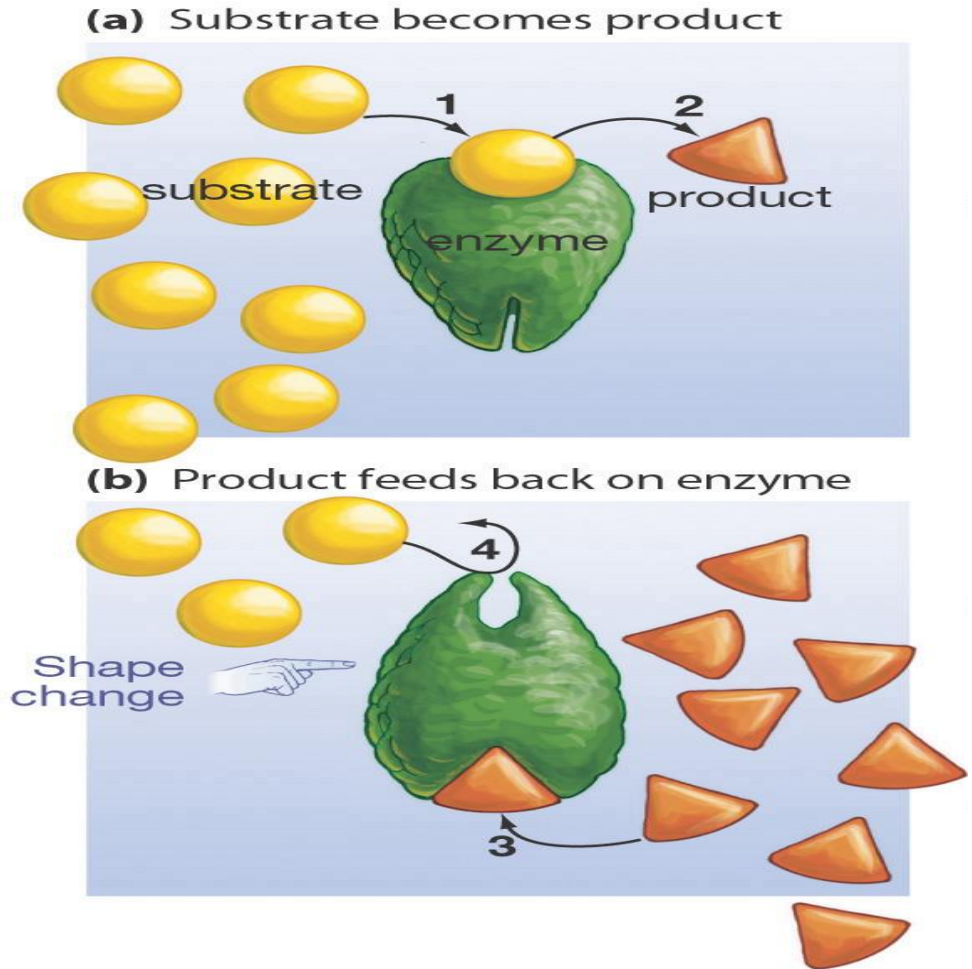
Inhibition

leading to

Changes in $V_{\max}$ or affinity for substrate K_m , or both

3. Feedback inhibition

-Feedback inhibition
a common form of
enzyme regulation in
which the product
inhibits the enzyme .



4.Zymogens (Proenzymes)

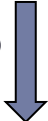
- ▶ A zymogen is an enzyme produced in an inactive form.
- ▶ It is converted to its active form when needed by removing a part of the zymogen.
- ▶ E.g. of zymogens are the digestive enzymes.

5. Covalent modification

- ▶ **Adding/removing groups** – like phosphate groups

Phosphorylation and de-phosphorylation

e.g.:

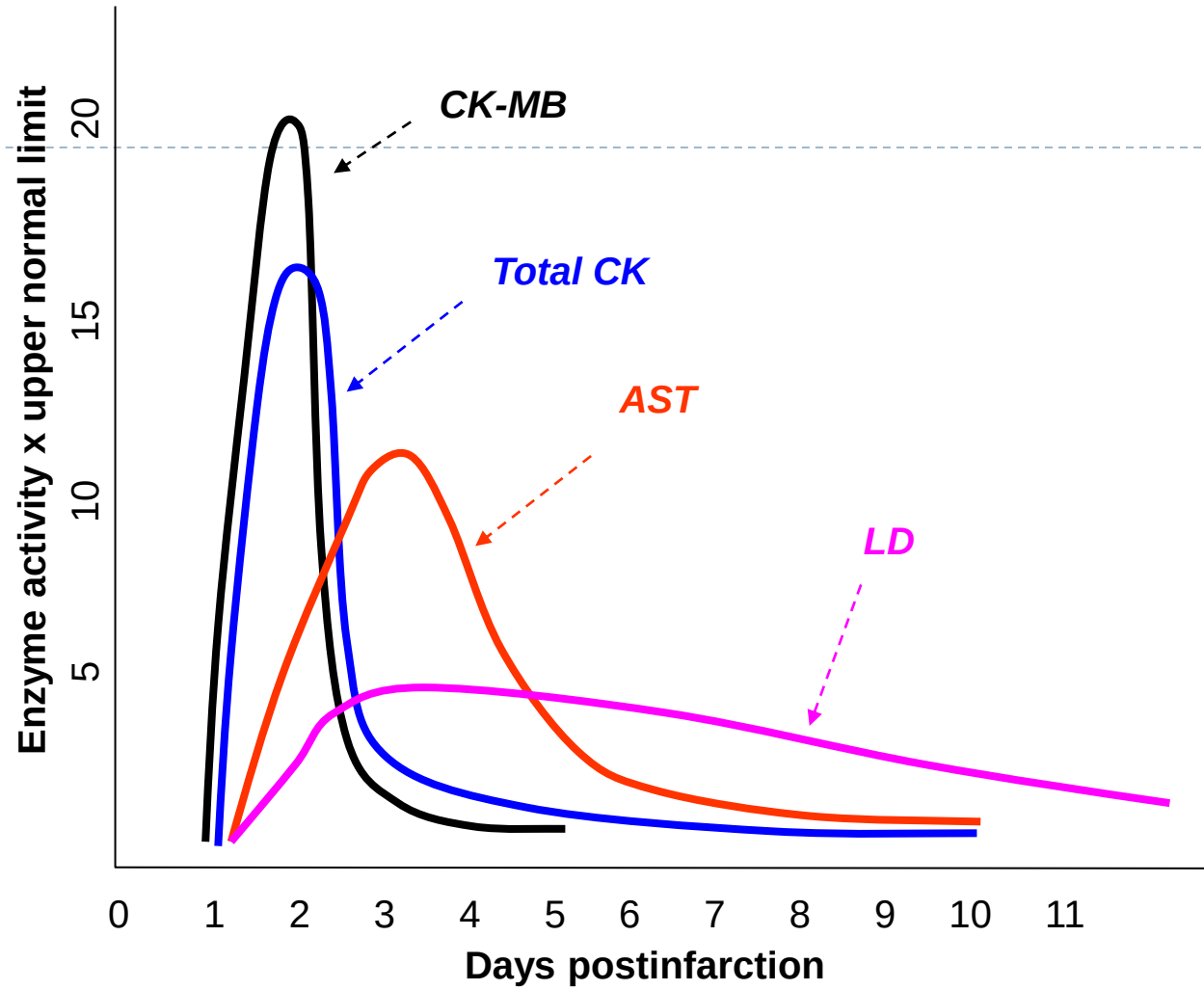
Glycogen synthase + P  ACTIVITY

glycogen phosphorylase + P  ACTIVITY

Diagnostic value of enzymes

- ▶ Enzymes may be measured in cells, such as red blood cells, white blood cells and other tissue cells to detect enzyme deficiency in certain diseases e.g. deficiency of red cell glucose 6-phosphate dehydrogenase in favism.
- ▶ Some enzymes show relatively high activity in only one or a few tissues. The presence of increased levels of these enzymes in plasma thus reflects damage to the corresponding tissue. For example, the enzyme alanine aminotransferase (ALT) is abundant in the liver.

-
- ▶ Troponin I & II is the earliest marker of myocardial infarction. Enzymes include creatine kinase, aspartate transaminase and lactate dehydrogenase. The temporal changes of serum enzyme levels after an attack of acute myocardial infarction is shown .



ISOENZYMES

- ▶ Isoenzymes or isozymes, are enzymes that **catalyze the same reaction** but **differ in their structure** (amino acid sequence), **properties** (K_m affinity to substrate, heat stability, susceptibility to inhibition, and electrophoretic mobility).
- ▶ They are usually produced by different cells or by different compartments of the same cell in the same species. Examples of isoenzymes present in humans are lactate dehydrogenase, creatine kinase and alkaline phosphatase

-
- ▶ For example, creatine kinase occurs as three isoenzymes. Each isoenzyme is a dimer composed of two polypeptides (called B and M subunits) associated in one of three combinations:
 - ▶ CkI = BB,
 - ▶ CK2 = MB, and
 - ▶ CK3 = MM.



~ End ~